

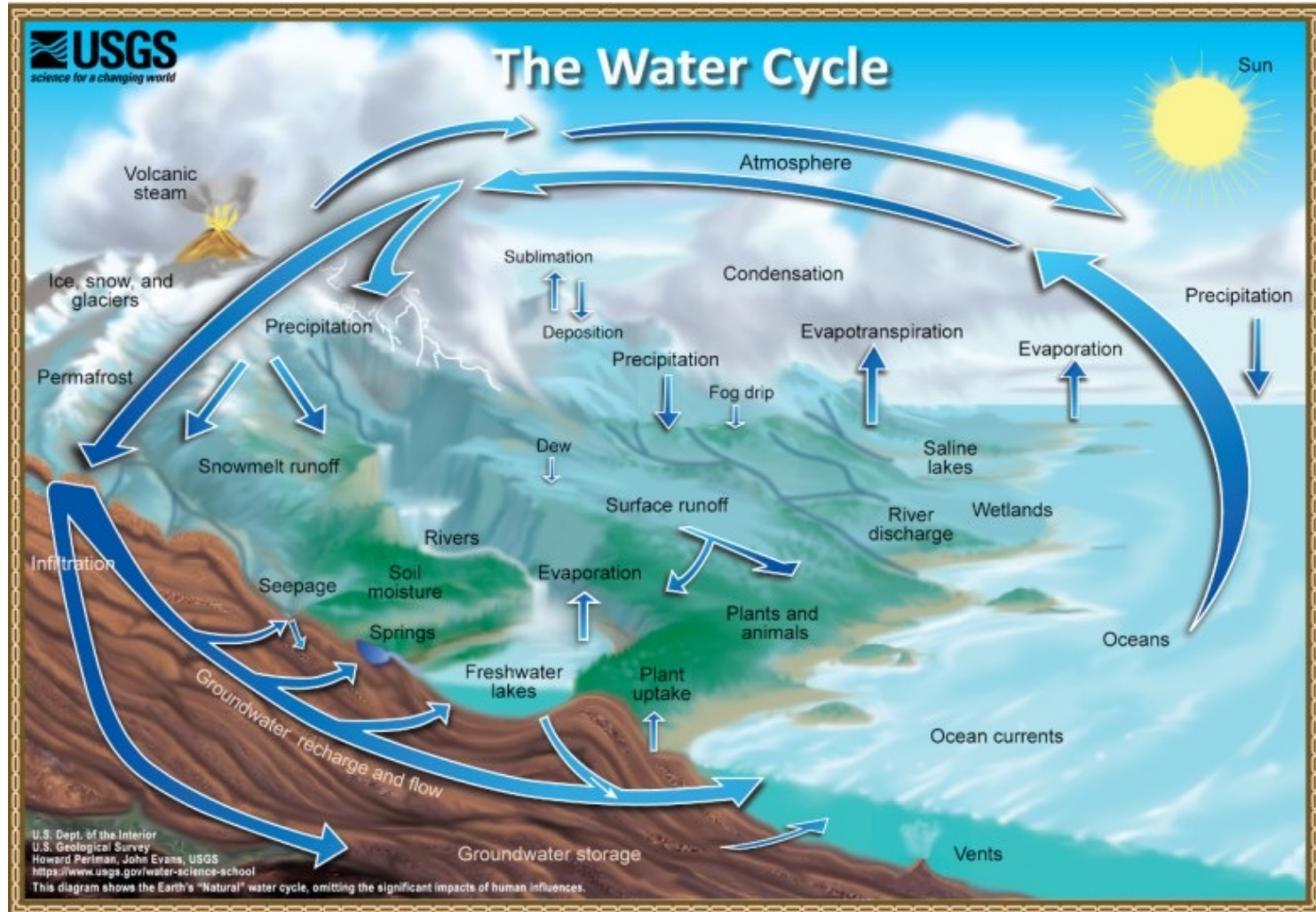
Lecture 04

Extreme Event Analysis-Part I (GEV Distribution)

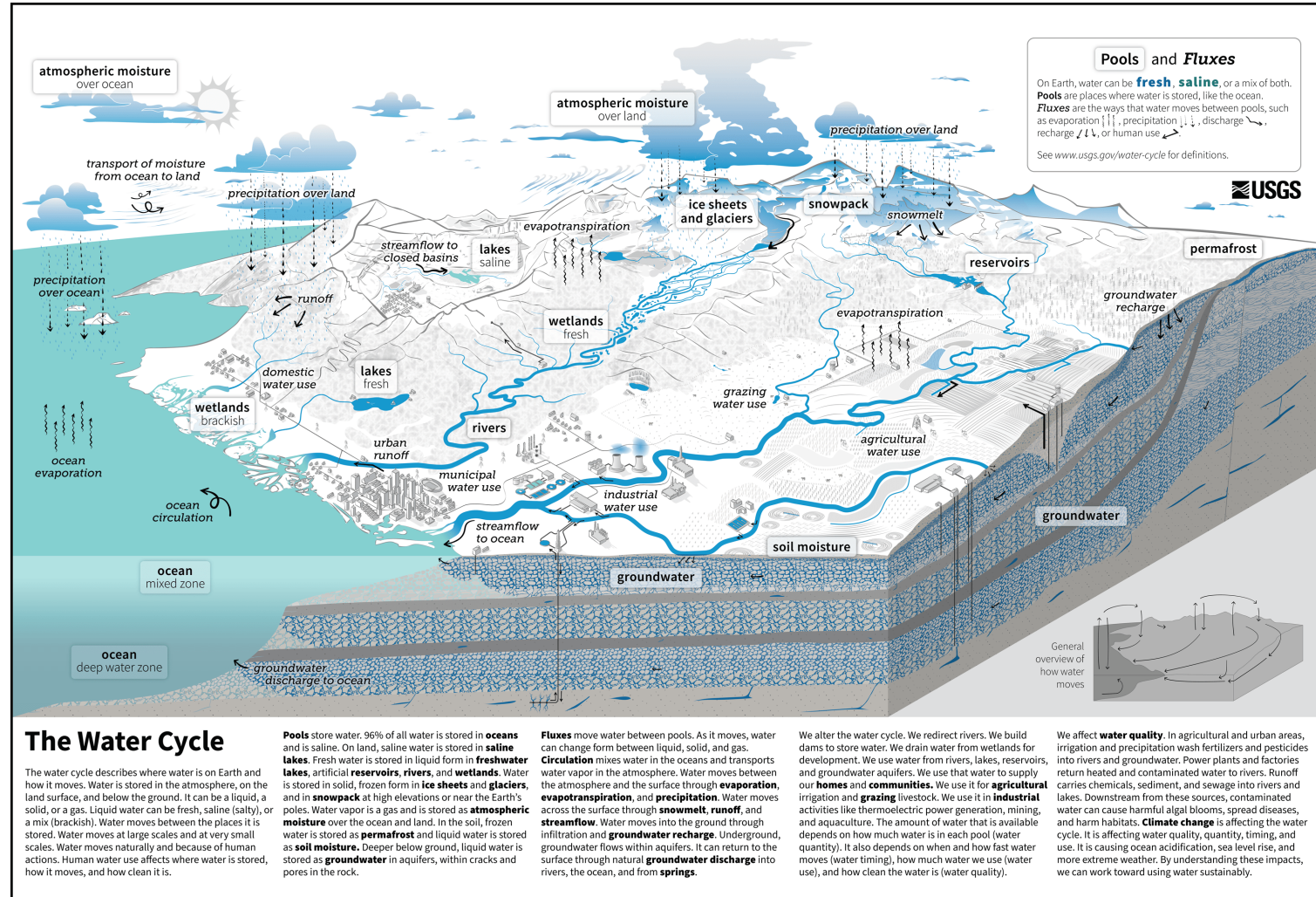
한강홍수통제소 방문



Old Water Cycle Diagram (USGS, 2000)



New Water Cycle Diagram (USGS, 2022)



Review on Lecture 3

❖ Steps for Probabilistic Modeling

- ◆ Step 1: Adopt family of probability models:
Normal, Exponential, Gamma, Binomial, Poisson, ...
- ◆ Step 2: Fit the model to data (Estimate Parameters)

❖ Corresponding Questions to be Answered

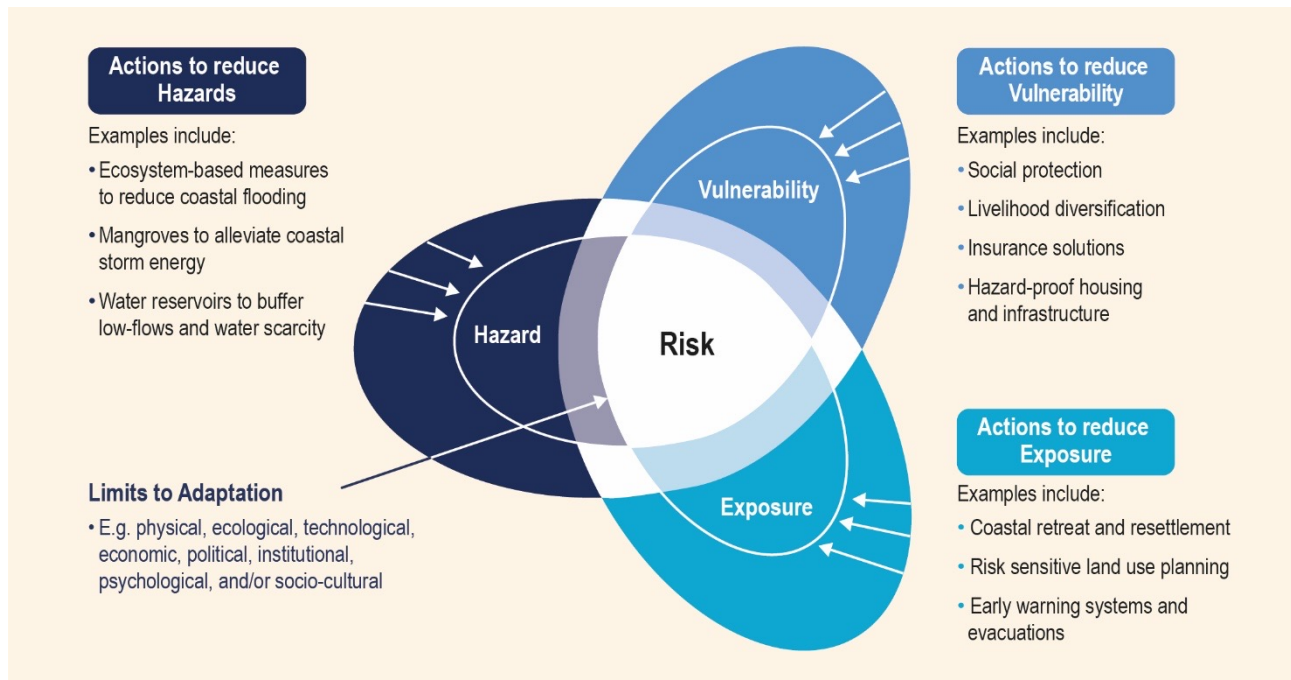
- ◆ Step 1: How to choose the appropriate model?
 - [1] **Physical Basis** (물리적인 특성을 바탕으로 선택)
 - [2] **Limit Laws** (극한 정리) Central Limit Theorem
 - [3] **Empirical Basis**: Which model fits data best? AIC/P-P or Q-Q/ χ^2
- ◆ Step 2: How do we estimate parameters?
 - Maximum Likelihood Estimation (MLE)

Today's Topic: Lecture 04

- ❖ Extreme Event Analysis: Focusing on Floods!
- ❖ Questions to be answered in today's lecture:
 - ◆ What distribution to use?
 - ◆ What design event to use?
- ❖ Distributions for Extreme Values
 - ◆ Three Types of Extreme Distributions
 - Gumbel, Fréchet, Weibull (Type 1, Type II, Type III)
 - ◆ Generalized Extreme Value (GEV) Distribution

Definitions & Notations

- ❖ Flood Hazard: flood magnitudes & its likelihood of occurrence
- ❖ Design Level: magnitude of the event to be considered for design of project



Definitions & Notations

- ❖ Flood Hazard: flood magnitudes & its likelihood of occurrence
- ❖ Design Level: magnitude of the event to be considered for design of project
- ❖ Return Period: suppose X exceeds some level X_T with probability $1/T$; $P(X > X_T) = 1/T$

- ❖ Commonly used notations
 - ◆ Non-exceedance probability
 - ◆ Exceedance probability
 - ◆ Equivalent Design Events

Exceedance Probability Example

❖ Let X be annual maximum streamflow the 100-year flood will be exceeded with probability in any given year:

◆ On average, $X > X_{100}$ every 100 years, but X_{100} can be exceeded more than once or not at all in a 100 year span.

[Q] What is the probability that X_{100} is exceeded at least once during the 30-year period?

Questions regarding Return Periods

- ❖ For a specific type of structure, what should T be?
- ❖ Once we do choose T , how should we find X_T ?
- ❖ Does this approach apply on all cases?
 - ◆ NO!
 - ◆ If "Safety" is a big concern, we design to "upper limit" hazard, such as:
 - Probable Maximum Precipitation (PMP)
 - Probable Maximum Flood (PMF)

A1) Return Periods for ...

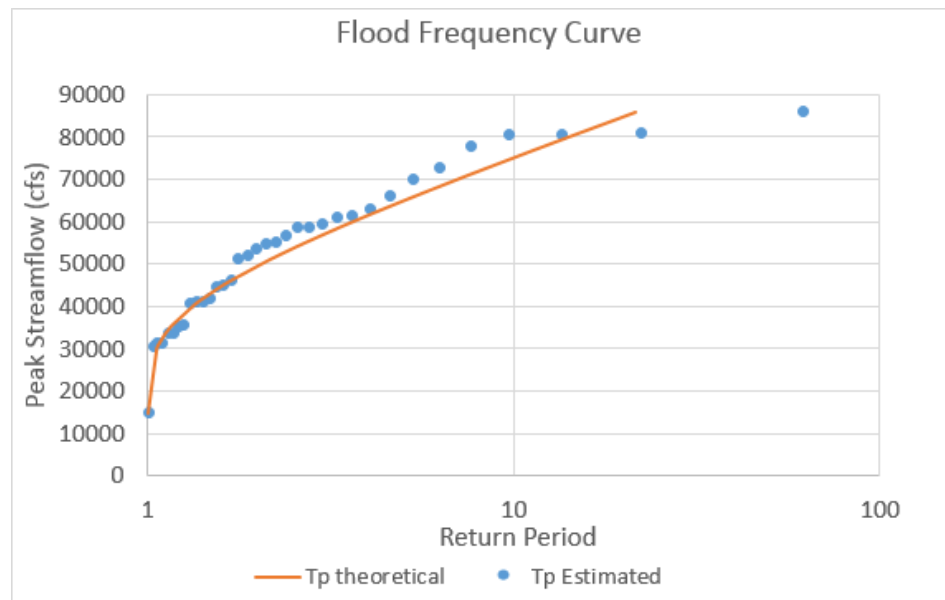
구 조 물 종 류		설계빈도(년)
배수시설		2~50
	배수로	20 이상
	방수로	20 이상
	배수제	20 이상
	배수문	20 이상
	배수펌프	20 이상
	유수지 및 저류지	20 이상
하천제방	국가 하천	100~200
	지방 하천	50~200
홍수방어(조절)용	저수지	50~SPF(표준설계홍수량)
	여수로	PMF(가능최대홍수량)
	제 방	10~SPF(표준설계홍수량)

출처: 하천설계기준해설

Calculating Extreme Events

❖ Extreme Value Theory (극단치 이론)

- ◆ Goal: accurately estimation of design events (extreme values) based on a ***limited set*** of observations
- ◆ Can we use a single distribution to fit all the data & examine?
 - No, because the data decays faster at the tail than observed.



Calculating Extreme Events

- ❖ Block maxima

- ◆ We are interested in the statistical behavior of Z_n where

Calculating Extreme Events

❖ What is $F_Z(z) = P(Z \leq z)$ (in case of independence)?

◆ Strategy #1:

Estimate $F_X(z|\theta)$ with $\hat{\theta}$, then calculate $F_Z(z)$ as

Calculating Extreme Events

❖ What is $F_Z(z) = P(Z \leq z)$ (in case of independence)?

◆ Strategy #2: Look for family of models for $F_Z(z)$ directly.

- Analogy:

*** Extreme Value Theorem (EVT) ***

Part II: Extreme Value Distributions

❖ $Z_n = \max(X_1, X_2, \dots, X_n)$ 에 대해서, Extreme Value Theorem에 의해서 $F_Z(z)$ 는 아래 세 가지 분포 중 하나를 따른다:

(Haan Chapter 6, p.132~)

- ◆ Gumbel (Type I)
- ◆ Fréchet (Type II)
- ◆ Weibull (Type III)

Part II: Extreme Value Distributions

❖ Gumbel (Type I)

The probability density function for the type I extreme value distribution is

$$p_X(x) = \frac{1}{\alpha} \exp\left(-\frac{x - \beta}{\alpha} - \exp\left[-\frac{x - \beta}{\alpha}\right]\right)$$
$$-\infty < X < \infty; -\infty < \beta < \infty; \alpha > 0 \quad (6.47)$$

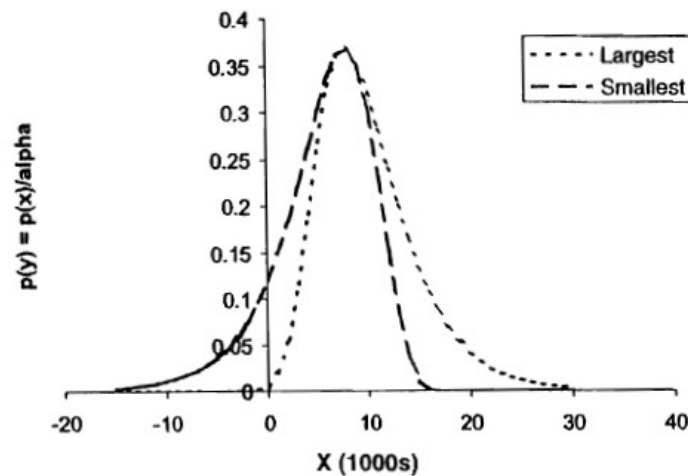


Fig. 6.7. Example of extreme value type I density curves.

Part II: Extreme Value Distributions

❖ Fréchet (Type II)

Part II: Extreme Value Distributions

❖ Weibull (Type III)

distribution of low stream flows. Naturally low flows are bounded by zero on the left. The type III for minimum values is also known as the Weibull distribution and is defined as

$$p_X(x) = \alpha x^{\alpha-1} \beta^{-\alpha} \exp[-(x/\beta)^\alpha] \quad X \geq 0; \alpha, \beta > 0 \quad (6.60)$$

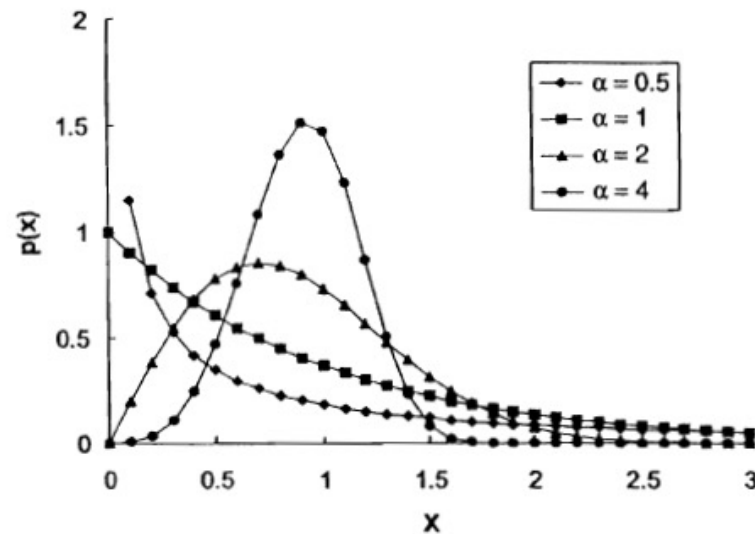


Fig. 6.8. Examples of extreme value type III minimum (Weibull) density curves.

❖ Fitting Strategy

- ◆ Pick a model: Type I, Type II, Type III
- ◆ Fit the parameters of the model via MLE

❖ Arising Problems from the above strategy:

- ◆ How should we pick the model?
- ◆ Once the decision is made, uncertainty in that choice cannot be considered afterwards.

❖ Solution:

Generalized Extreme Value (GEV)

GENERALIZED EXTREME VALUE DISTRIBUTION

Stedinger et al. (1994) discuss the generalized extreme value (GEV) distribution given by

$$P_X(x) = \exp\left(-\left[1 - \frac{\kappa(x - \xi)}{\alpha}\right]^{\frac{1}{\kappa}}\right) \quad \text{for } \kappa \neq 0 \quad (6.76)$$



Other Distributions for Extreme Values

❖ Beta Distribution (Haan p.140)

BETA DISTRIBUTION

A distribution that has both an upper and lower bound is the beta distribution. Generally, the beta distribution is defined over the interval 0 to 1. It can, however, be transformed to any interval α to β . If the limits of the distribution are unknown, they become parameters of the distribution, making it a 4-parameter rather than a 2-parameter distribution. The beta density function is given by

$$p_X(x) = \frac{x^{\alpha-1}(1-x)^{\beta-1}}{B(\alpha, \beta)} \quad 0 < x < 1; \alpha, \beta > 0 \quad (6.86)$$

The function $B(\alpha, \beta) = \int_0^1 x^{\alpha-1}(1-x)^{\beta-1} dx$ is called the beta function. The beta function is related to the gamma function by

$$B(\alpha, \beta) = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha + \beta)} \quad (6.87)$$

Other Distributions for Extreme Values

❖ Pearson Distribution (Haan p.141)

PEARSON DISTRIBUTIONS

Karl Pearson (Elderton 1953) has proposed that frequency distributions can be represented by

$$p_X(x) = \exp \left[\int_{-\infty}^x \frac{t + \alpha}{\beta_0 + \beta_1 t + \beta_2 t^2 + \dots} dt \right] \quad (6.90)$$

By choosing appropriate values for the parameters, equation 6.90 becomes a large number of families of distributions including the normal, beta, and gamma distributions.

The Pearson type III has found application in hydrology especially as the distribution of logarithms of flood peaks. This distribution can be written

$$p_X(x) = p_0 \left(1 + \frac{x}{\alpha} \right)^{\alpha/\delta} e^{-x/\delta} \quad (6.91)$$

(Mid) Term Project Report

In Teams, or Individual?

- ❖ 본 수업에서는 매개변수 추정을 위해 주로 MLE 방법을 소개하였지만, Method of Moments (Haan p.72), Method of L-moments (Handbook of Hydrology 18.1.5), Bayesian inference 등의 다양한 방법을 사용해도 매개변수 추정이 가능합니다.
 - ◆ 위의 방법들에 대한 소개(설명)을 간단히 진행해 주세요.
기본 개념, 매개변수 추정 시 사용되는 수식, L-moment Diagram과 같은 그림을 해석하는 방법 등으로 설명을 진행하면 되겠습니다.
 - ◆ 실제 데이터를 사용하여 각 방법들에 대한 비교를 진행해 보세요.
보통 매개변수 추정을 위해 어떤 방법을 많이 사용하는지,
그리고 각 방법들의 장단점은 무엇인가요?
 - ◆ 실제 데이터는 WAMIS나 MyWater를 통해 일별 강수량/유입량 자료를 다운로드 받아 사용해서 비교를 진행하면 됩니다.

(Mid) Term Project Report

(예습 프로젝트-개인별로 진행)

- ❖ 미국에서는 홍수 빈도 분석 (**Flood Frequency Analysis** or **Flood Flow Frequency**)을 위해 Bulletin 17B (or 17C)에 마련되어 있는 가이드라인을 사용합니다. 간략하게 어떤 절차를 통해서 홍수 빈도 분석을 진행하는지 조사해 봅시다.
- ❖ 이에 반해, 우리나라에서는 어떤 방법으로 홍수 빈도 분석을 진행하고 있나요? 위의 미국의 방법과는 어떤 공통점 및 차이점이 있는지 비교 분석을 진행해 주시기 바랍니다.
- ◆ 두 개의 프로젝트를 합해 총 5페이지를 넘지 않도록 (그림 포함) 작성해 주시기 바랍니다. 11월 1일 화요일 수업에서 리포트를 가지고 팀별/개인별로 짧게 발표하는 시간을 가지려고 하니, 가독성 있게 준비해 주세요.

Future Schedule Preview

❖ 다음 주 강의 (2022-10-25 화요일 18:30-21:20)

◆ **Lec 05: Extreme Event Analysis-Part II**

- Flood Frequency Analysis using LP3 Distributions
- Introduction to Peaks-Over-Threshold (POT) Models

❖ Other Assignments (Plans for October)

◆ Datacamp Assignments (To be assigned)

- Introduction to Tidyverse due 10/28
- Cleaning Data in R due 11/04

◆ Mid-Term Project Report & Presentation: 11/01 Tue